Sri Lanka Institute of Information Technology

Information Technolog

Trigonometry

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Year 01 and Semester 01

Outline

1. Angles
2. Trigonometric Identities
3. Trigonometric Formulas

Angles

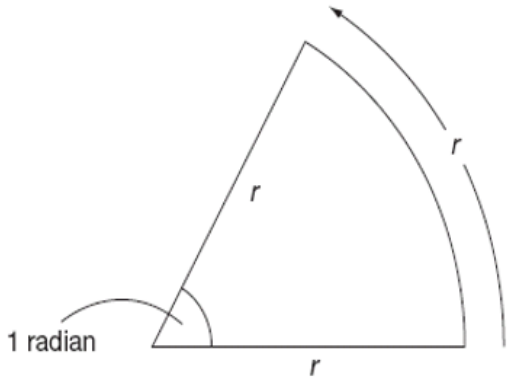
- Trigonometric identities
- Trigonometric formulas

Angles: Rotation

- When a straight line is rotated about a point, it sweeps out an angle that can be measured in degrees or radians.
- A straight line rotating through a full angle and returning to its starting point is said to have rotated through 360 degrees (360°).
- One degree = 60 minutes and one minute = 60 seconds.

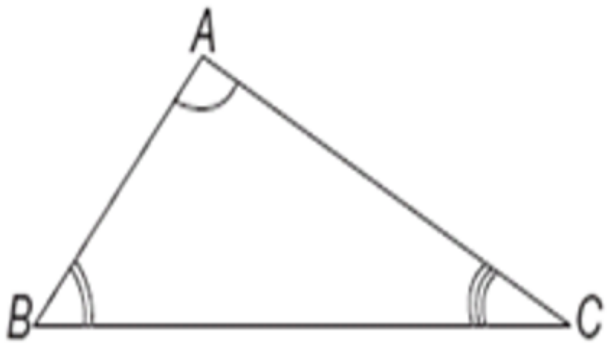
Angles: Radians

- When a straight line of length r is rotated about one end so that the other end describes an arc of length r, the line is said to have rotated through 1 radian (1 rad).



Angles: Triangles

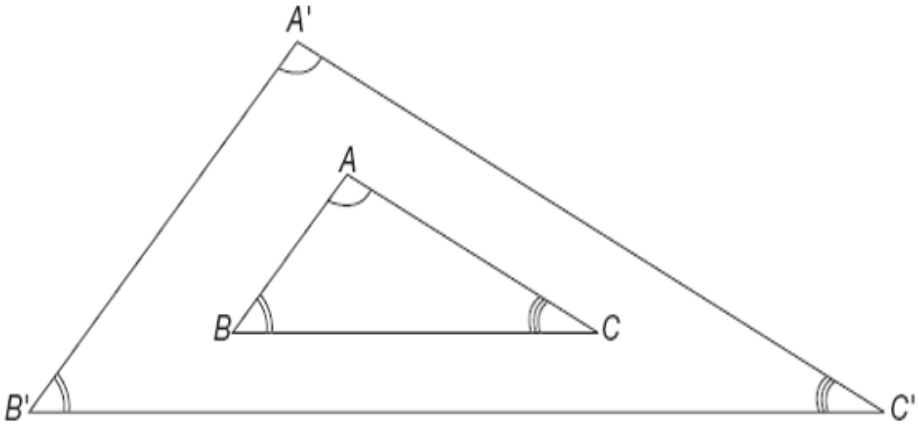
- All triangles possess shape and size.
- The shape of a triangle is governed by the **three angles** and the size by the **lengths of the three sides**.



Angles: Trigonometric Ratios

For similar triangles

$$\frac{AB}{A'B'} = \frac{AC}{A'C'} = \frac{BC}{B'C'}$$



so that:

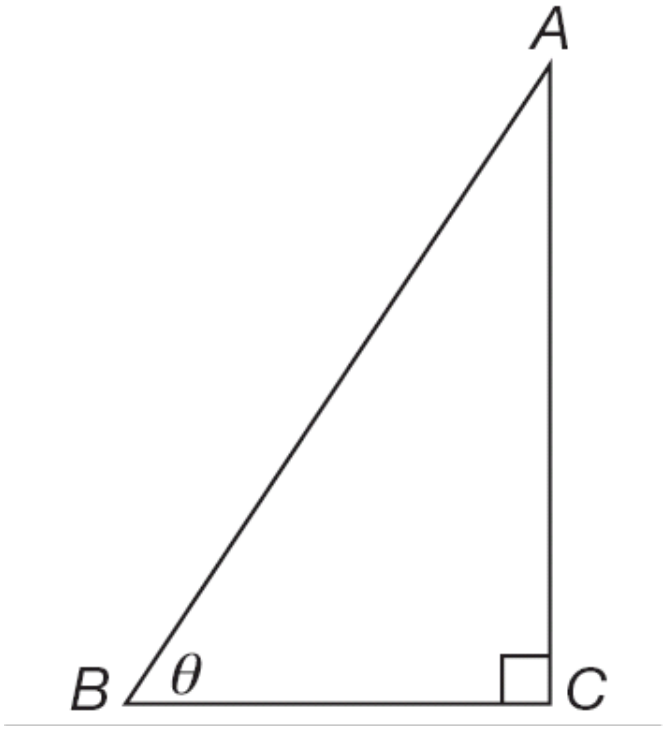
$$\frac{AB}{AC} = \frac{A'B'}{A'C'} \quad \text{and} \quad \frac{AB}{BC} = \frac{A'B'}{B'C'} \quad \text{and} \quad \frac{AC}{BC} = \frac{A'C'}{B'C'}$$

Angles: Trigonometric Ratios

$$\textit{sine} \text{ of angle } \theta = \frac{AC}{AB} \text{ - denoted by } \sin\theta$$

$$\textit{cosine} \text{ of angle } \theta = \frac{AC}{BC} \text{ - denoted by } \cos\theta$$

$$\textit{tangent} \text{ of angle } \theta = \frac{AC}{BC} \text{ - denoted by } \tan\theta$$



Angles: Reciprocal Ratios

$$\textit{cosecant} \text{ of angle } \theta = \frac{1}{\sin\theta} \text{ - denoted by } \operatorname{cosec}\theta$$

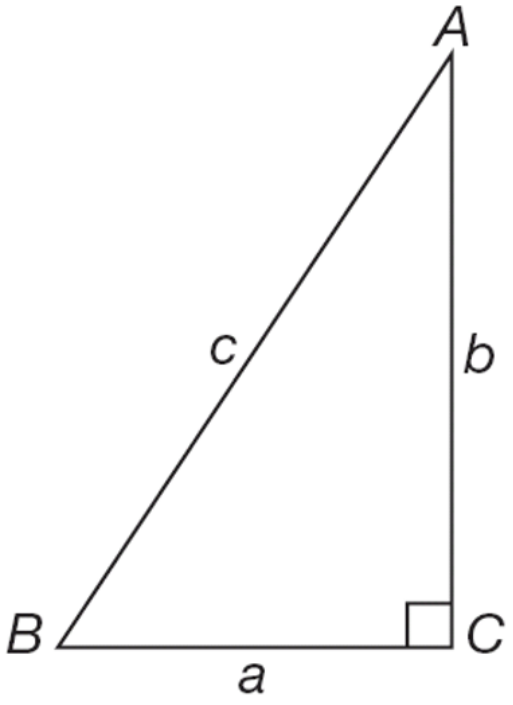
$$\textit{secant} \text{ of angle } \theta = \frac{1}{\cos\theta} \text{ - denoted by } \sec\theta$$

$$\textit{cotangent} \text{ of angle } \theta = \frac{1}{\tan\theta} \text{ - denoted by } \cot\theta$$

Angles: Pythagoras' Theorem

- The square on the hypotenuse of a right-angled triangle is equal to the sum of the squares on the other two sides.

$$a^2 + b^2 = c^2$$



Angles: Special Triangles

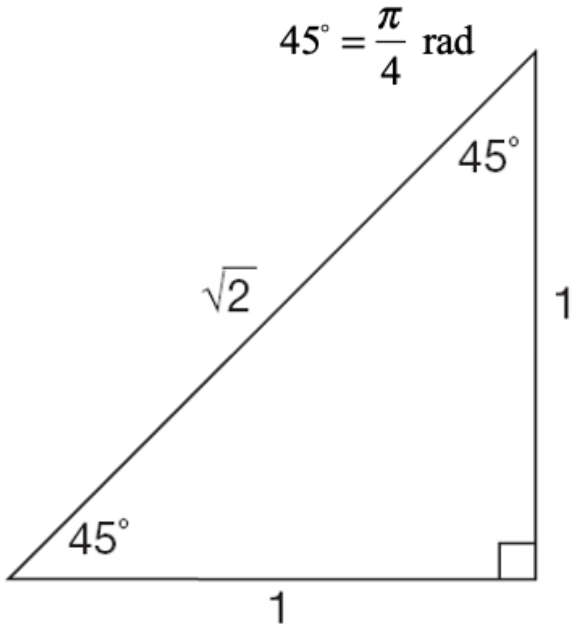
Right-angled isosceles

Angles measured in degrees:

$$\sin 45^\circ = \cos 45^\circ = \frac{1}{\sqrt{2}} \quad \text{and} \quad \tan 45^\circ = 1$$

Angles measured in radians:

$$\sin \pi/4 = \cos \pi/4 = \frac{1}{\sqrt{2}} \quad \text{and} \quad \tan \pi/4 = 1$$



Angles: Special Triangles

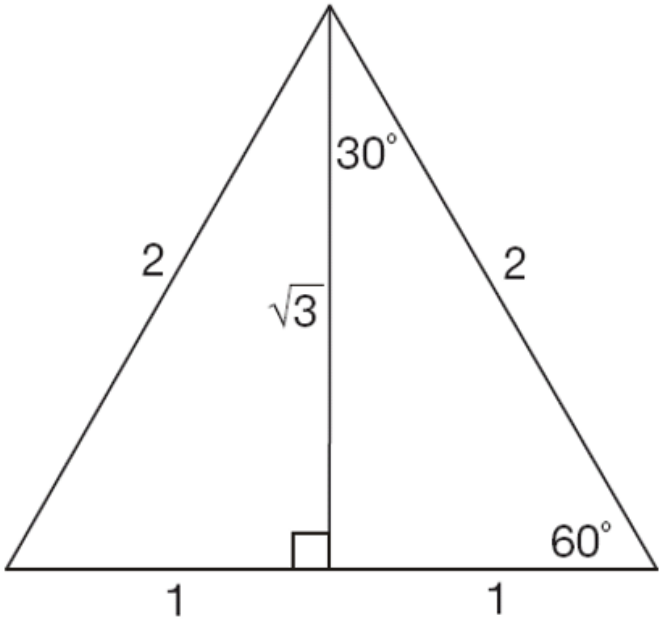
Half equilateral

Angles measured in degrees:

$$\sin 30^\circ = \cos 60^\circ = \frac{1}{2}$$

$$\sin 60^\circ = \cos 30^\circ = \frac{\sqrt{3}}{2}$$

$$\tan 60^\circ = \frac{1}{\tan 30^\circ} = \sqrt{3}$$



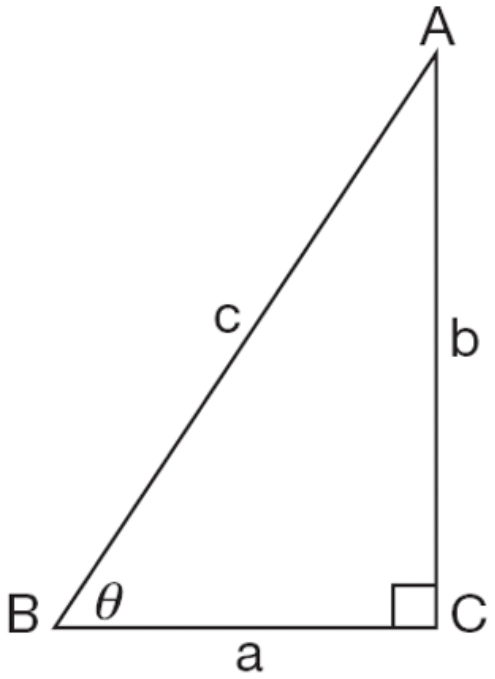
Trigonometric Identities: The fundamental identity

- The fundamental trigonometric identity is derived from Pythagoras' theorem.

$$a^2 + b^2 = c^2 \quad \text{so} \quad \frac{a^2}{c^2} + \frac{b^2}{c^2} = 1$$

that is:

$$\boxed{\cos^2\theta + \sin^2\theta = 1}$$



Trigonometric Identities: Two more identities

- Dividing the fundamental identity by $\cos^2\theta$

$$\cos^2\theta + \sin^2\theta = 1 \quad \text{so that} \quad \frac{\cos^2\theta}{\cos^2\theta} + \frac{\sin^2\theta}{\cos^2\theta} = \frac{1}{\cos^2\theta}$$

that is:

$$\boxed{1 + \tan^2\theta = \sec^2\theta}$$

Trigonometric Identities: Two more identities

- Dividing the fundamental identity by $\sin^2\theta$

$$\cos^2\theta + \sin^2\theta = 1 \quad \text{so that} \quad \frac{\cos^2\theta}{\sin^2\theta} + \frac{\sin^2\theta}{\sin^2\theta} = \frac{1}{\sin^2\theta}$$

that is:

$$\boxed{\cot^2\theta + 1 = \operatorname{cosec}^2\theta}$$

Trigonometric Formulas

- Sums and differences of angles

$$\cos(\theta+\phi)=\cos\theta\cos\phi-\sin\theta\sin\phi$$
$$\cos(\theta-\phi)=\cos\theta\cos\phi+\sin\theta\sin\phi$$

$$\sin(\theta+\phi)=\sin\theta\cos\phi+\cos\theta\sin\phi$$
$$\sin(\theta-\phi)=\sin\theta\cos\phi-\cos\theta\sin\phi$$

$$\tan(\theta+\phi)=\frac{\tan\theta+\tan\phi}{1-\tan\theta\tan\phi}$$
$$\tan(\theta-\phi)=\frac{\tan\theta-\tan\phi}{1+\tan\theta\tan\phi}$$

Trigonometric Formulas

- Double angles

$$\cos 2\theta = \cos^2\theta - \sin^2\theta$$
$$\cos 2\theta = 1 - 2\sin^2\theta$$
$$\cos 2\theta = 2\cos^2\theta - 1$$

$$\sin 2\theta = 2\sin\theta\cos\theta$$

$$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$$

Trigonometric Formulas

- Sums and differences of ratios

$$\sin\theta + \sin\phi = 2\sin\frac{\theta+\phi}{2}\cos\frac{\theta-\phi}{2}$$

$$\sin\theta - \sin\phi = 2\cos\frac{\theta+\phi}{2}\sin\frac{\theta-\phi}{2}$$

$$\cos\theta + \cos\phi = 2\cos\frac{\theta+\phi}{2}\cos\frac{\theta-\phi}{2}$$

$$\cos\theta - \cos\phi = -2\sin\frac{\theta+\phi}{2}\sin\frac{\theta-\phi}{2}$$

Trigonometric Formulas

- Products of ratios

$$2\sin\theta\cos\phi = \sin(\theta+\phi) + \sin(\theta-\phi)$$

$$2\cos\theta\cos\phi = \cos(\theta+\phi) + \cos(\theta-\phi)$$

$$2\sin\theta\sin\phi = \cos(\theta-\phi) - \cos(\theta+\phi)$$

Thank You!